

The 2025 GCPC Collegiate Programming Contest

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difficulty: easy

Time limit per test: 1s

Problem Childhood

In a playground, there are children standing in a row. Each child has a ball and throws it in a particular direction. Uchiha observed these children playing and recorded the position of each child along with the distance they threw their ball. Now he wants to know if there are two children who threw their balls at each other.

The trajectory of a child's ball is a straight line: if a child at position x_i throws a ball d_i meters to the right, the ball will land at position $x_i + d_i$. If the distance d_i is negative, the ball is thrown to the left, landing at position $x_i - |d_i|$.

Input

The first line contains an integer n ($1 \leq n \leq 50$) — the number of children in the playground. The next n lines each contain two integers x_i and d_i ($-10^8 \leq x_i \leq 10^8$, $1 \leq |d_i| \leq 2 \cdot 10^5$) — representing the position x_i of the i -th child, and the distance d_i that the i -th child threw their ball.

Output

If there are two children such that child i threw a ball to child j and child j threw their ball back to child i , print "YES". Otherwise, print "NO".

Example

Input:	Output:
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3 0 1 1 1 2 -2	NO
2 0 2 2 -2	YES